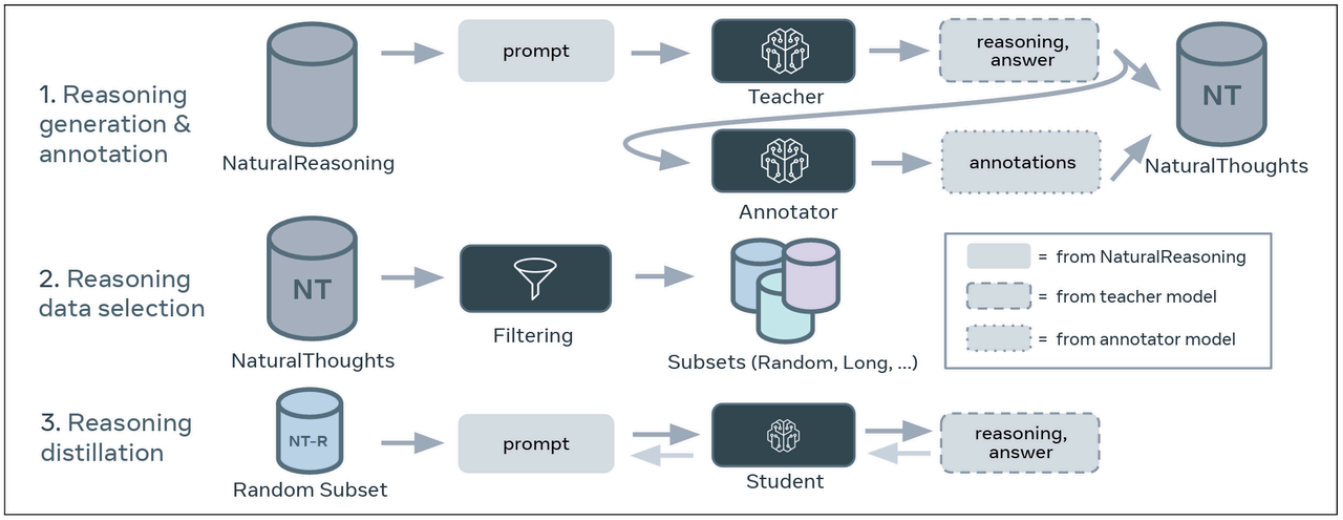
NaturalThoughts: Distilling General Reasoning via Strategic Trace Selection

> #Distillation #Thinking #Inference #Dataset #Finetune



- What is this paper about?
  - This project concentrates on distillation. The authors created a Dataset called Natural Thoughts and explored how selection and scaling affects the effectiveness of distillation.
- NaturalThoughts
  - A distillation dataset.
  - From the *NaturalReasoning* dataset, select various questions from various areas.
  - Let Deepseek-R1 form two parts of answers:
    - System 1: thinking process (CoT)
    - System 2: a short direct answer
  - Use an annotator model to label the function of some parts of CoT (*for example, reflections, corrections, self-verifications...*) and provide a mark for the **Verbosity** of the CoT (有多啰嗦？)
  - Selection
    - Diversity
      - Topics: uniformally from all areas
      - Question semantic embeddings
      - Reasoning Strategies: Need diverse solving strategies yet no overthinking
      - **Area diversity < Reasoning strategy diversities**: more diverse in solving strategy = better data point
    - Difficulty
      - Length and Verbosity: medium length = better
      - Model Agreement: if a larger thinking model can solve but not for a small non-thinking model, then that's a hard problem. Hard problem = better distillation.
  - Mode of Dilstillation

        - System-2 distillation: entire teacher inference: CoT+direct answer
        - System-1 distillation: final answer ONLY
        - Mixed system-1 and system-2: sampling of these two types (better)
        - Adaptive thinking:
            - No think: EXPLITELY tell the model no thinking needed
            - Think: EXPLITELY tell the model to think by adding `"Think carefully before answering. Use about {K} words" and a <think> token`
            - Adaptive-Think: tell the model to "think carefully" but without a `<think>` token.
- Key findings:
    - Scaling: larger # of data = better, challenging the current "less is more"
    - Providing a selection strategy: diverse reasoning strategy + model disagreement (hard) can be better
    - Answer & CoT+Answer mix = a balance between inference cost and correctness
- My Thoughts
    - The author's "adaptive thinking" remains too passive. Could we train a "thinking router" based on input confusion levels? It would automatically determine how many tokens a question requires for processing, rather than relying on user prompts.
    - This provides a good methodology: when distilling models, try to find problems w/ multiple reasoning strategies + hard in difficulty.
- Work Cited
    - Paper
    - Chat History